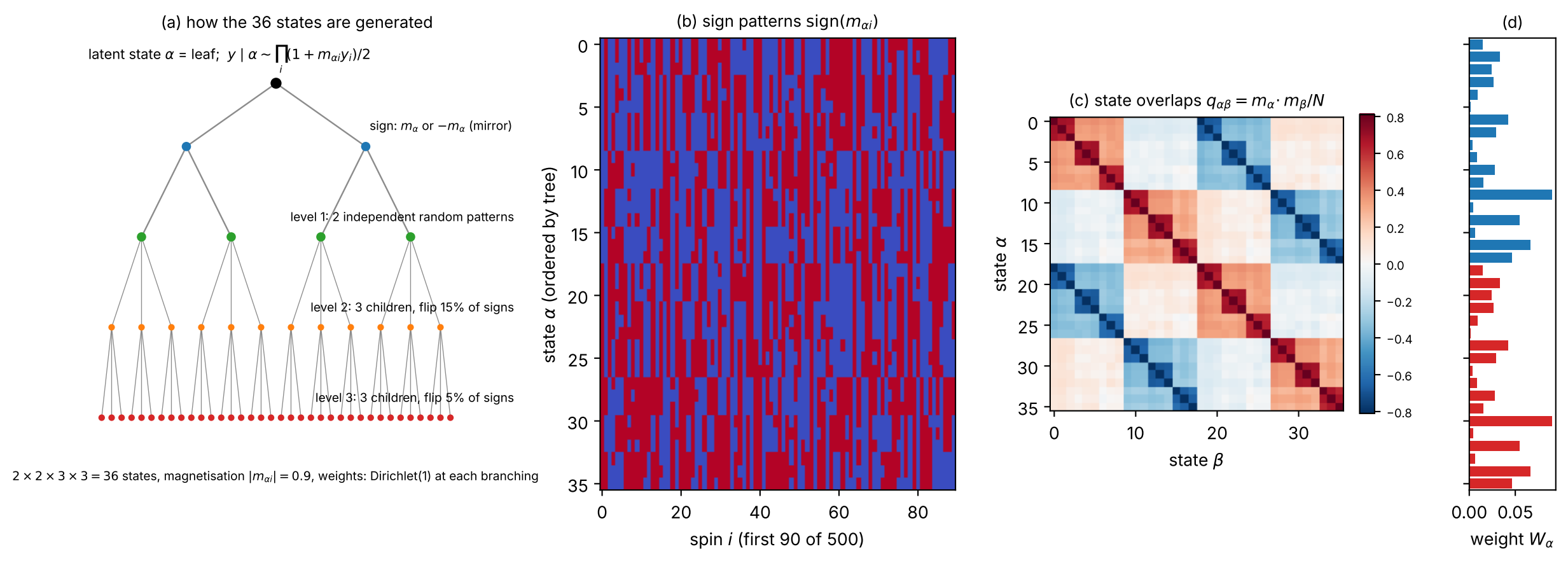
(a) how the 36 states are generated
latent state α = leaf; $y \mid \alpha \sim \prod_i (1 + m_{\alpha i} y_i)/2$
sign: m_α or $-m_\alpha$ (mirror)
level 1: 2 independent random patterns
level 2: 3 children, flip 15% of signs
level 3: 3 children, flip 5% of signs
$2 \times 2 \times 3 \times 3 = 36$ states, magnetisation $|m_{\alpha i}| = 0.9$, weights: Dirichlet(1) at each branching
(b) sign patterns sign($m_{\alpha i}$)
state α (ordered by tree)
spin i (first 90 of 500)
(c) state overlaps $q_{\alpha\beta} = m_\alpha \cdot m_\beta / N$
state α
state β
(d)
weight W_α